

Sidan Lu

Department of Civil and Environmental Engineering, Rice University

E-mail: sl367@rice.edu

EDUCATION

Ph.D., Environmental Engineering, Louisiana State University	Aug 2017 – May 2021
M.S., Environmental Science, Beijing Normal University	Aug 2014 – May 2017
B.S., Environmental Engineering, Harbin Institute of Technology	Aug 2010 – May 2014

RESEARCH INTERESTS

Advanced water treatment; Soil remediation; Interfacial biochemical processes; Critical resource recovery

RESEARCH EXPERIENCE

Postdoctoral Associate

Rice University

Advisor: Qilin Li

Department of Civil and Environmental Engineering

Apr 2026 – Present

Investigate PFAS fate and migration using nanobubble and lab-on-a-chip technologies for soil remediation.

Postdoctoral Scholar

Yale University

Advisor: Julie B. Zimmerman, N.A.E.

School of the Environment

Sep 2024 – Feb 2026

Developed a one-step electrochemical route to convert lignin-derived catechol to the nylon precursor dimethyl adipate with an 87% product yield.

Postdoctoral Fellow

Princeton University

Advisor: Zhiyong Jason Ren

Department of Civil and Environmental Engineering

Jul 2021 – Aug 2024

Engineered electro-membrane for selective volatile fatty acid production from anaerobic electro-fermentation and achieved pure volatile fatty acid extraction from pH-neutral fermented wastewater.

Graduate Student Researcher

Louisiana State University

Advisor: Xiuping Zhu

Department of Civil and Environmental Engineering

Aug 2017 – May 2021

Dissertation: Resource and Energy Recovery from Wastewater Using Advanced Bioelectrochemical Systems.

Graduate Student Researcher

Beijing Normal University

Advisor: Yujiao Sun

Dissertation: Responses of water and sediment microbial communities to seasonal nitrogen pollution in the Feng River, China.

GRANT AND PROPOSAL WRITING EXPERIENCE

- IWA Technology Development for Advancing Sanitation Innovation (TDASI) project, Decentralized sanitation and on-site water reuse using electrochemical processes. Role: Conceptualization, prepared text, drafted and revised full proposal. Rice University.
- Southwest Airline (SWA) Grant Program, Lignin to jet-range aromatic hydrocarbons by an advanced electrochemical approach. Role: Conceptualization, prepared text and figures, drafted and revised full proposal. Yale University.
- ACS Green Chemistry Institute Grant Program, Early Career Postdoctoral-Faculty Bridge Grant, Sustainable electrocatalytic synthesis of aviation fuels from waste biomass. Role: Conceptualization, prepared text and figures, drafted and revised full proposal. Yale University.
- Rice Future Faculty Fellows (FFF) program. Development of novel materials and processes to enhance the energy efficiency and cost-effectiveness of water use and reuse. Role: Prepared text and figures, drafted and revised full proposal. Rice University.
- The ACS Younger Chemists Committee (YCC) and the CIBA Foundation, Energy-efficient electrochemical methods to convert biomass into nylon precursor with high yields. Role: Conceptualization and drafted and revised full proposal. Yale University.
- ACS Green Chemistry Institute Grant Program, Advanced electrochemical technologies for producing catechol as a platform chemical from biomass-derived feedstocks. Role: Conceptualization and drafted and revised full proposal. Yale University.
- The University of California President's Postdoctoral Fellowship Program. Addressing key challenges in water purification and resource management. Role: Conceptualization, prepared text and figures, drafted and revised full proposal. Yale University.

PROJECT EXPERIENCE

Decentralized Reuse Enabled by Autonomous Modules (DREAM)

2026–Present

Funding Agency: Gates Foundation Water, Sanitation, and Hygiene Program; Project INV-086263

Role: Postdoctoral Associate, Technical Lead, and Project Coordinator

- Build an automated modular treatment and reuse system for graywater and source-separated urine to support reliable decentralized operation and reduce treatment loads on reinvented toilet systems. This project includes a full transition from TRL5 to TRL6 through third-party engineering reviews, system fabrication, and field deployment in operational environments.
- Coordinate with Yokogawa and Intuitech on process design, P&IDs, bills of materials, controls integration, system operation, troubleshooting, sampling, and performance validation.

- Evaluate electrodialysis-based treatment for water recovery, energy demand, water quality, operating reliability, and scale-up readiness.

PFAS Fate and Transport in Soil Remediation

2026–Present

Funding Agency: Rice University

Role: Postdoctoral Associate and Technical Lead

- Investigate nanobubbles as mobile carriers for remediating PFAS-contaminated soil and evaluate how water chemistry and pore structure affect contaminant mobility.
- Develop lab-on-a-chip methods for in situ observation of PFAS fate and transport in porous media to support lower-cost soil-remediation technologies.

Electrochemical Conversion of Lignin to Jet-Range Aromatic Hydrocarbons

2025–2026

Funding Agency: Southwest Airlines

Role: Postdoctoral Fellow and Principal Investigator

- Secured project funding and led a five-researcher team advancing electrochemical conversion of lignin-derived compounds into sustainable aviation-fuel precursors.
- Defined research milestones, reviewed experimental data, coordinated technical work, and prepared sponsor deliverables and progress reports.

Sustainable Nylon Production from Lignin-Derived Catechol via Dimethyl Adipate

2024–2026

Funding Agency: The Heintzelman Carbon Sequestration Fund

Role: Postdoctoral Fellow and Technical Lead

- Overcame a major selectivity barrier and developed a one-step electrochemical route that converted lignin-derived catechol into dimethyl adipate, a nylon precursor, at 87% yield.
- Directed reaction development, product separation, analytical verification, and data interpretation.

Targeted extraction of valuable intermediate products and clean water from municipal wastewater using electroactive anaerobic membrane bioreactors (ANEMBR)

2021–2024

Funding agency: U.S. Department of Energy Advanced Materials and Manufacturing Technologies Office; Project DE-EE0009494

Role: Postdoctoral Fellow and Technical Lead

- Led a four-researcher team developing anaerobic electro-fermentation for selective volatile fatty acid production from wastewater and managed experimental plans, deliverables, and agency reports.
- Engineered temperature-controlled micro-redox conditions to improve product selectivity and evaluated process performance through chemical and microbial analyses.
- Developed membrane electrolysis distillation to recover purified volatile fatty acids from pH-neutral fermented wastewater by integrating pH control, electrochemical heating, and gas-liquid separation.

Advanced Microbial Electrochemical Systems for Space Wastewater Treatment

2017–2021

Funding Agency: NASA Louisiana Space Grant Consortium LaSPACE; Project GR-00004809

Role: Doctoral Researcher and Technical Lead

- Designed resource-recovery microbial fuel cells for urine-containing and space-wastewater treatment, including systems that operated without external energy consumption.
- Fabricated copper hexacyanoferrate electrodes and integrated them into a concentration flow cell enhanced by bioelectrochemical currents, achieving 42 W/m² energy recovery from salinity gradients.
- Synthesized molybdenum nitride nanobelt electrodes and developed electrochemical systems for low-cost hydrogen production and nutrient recovery.

HONORS AND AWARDS

- Conference Session Organization Recognition, 2026 Solid Waste Green Development Conference, China Environment Federation 2026
- Project Funding, Lignin to Jet-Range Aromatic Hydrocarbons by an Advanced Electrochemical Approach, Southwest Airlines 2025
- Oral Presentation Award, LSU Graduate Student Research Conference 2021
- Economic Development Award, Louisiana State University 2019
- Oral Presentation Award, 4th IWA Conference of Science Summit on Urban Water 2018
- Doctoral Student Scholarship, Louisiana State University 2017
- National Graduate Student Award 2016
- First-Class Scholarship, Beijing Normal University 2015
- First-Class Scholarship, Beijing Normal University 2014

PUBLICATIONS

Peer-Reviewed Publications

1. **Sidan Lu**, Fang Lin, Mahlet Garedew, Ho Yin Tse, Andrew T. Champlin, Julie B. Zimmerman, Hanno C. Erythropel*, and Paul T. Anastas*. One-Pot Electrochemical Synthesis of the Nylon Precursor Dimethyl Adipate from Lignin-Derived Catechol. *Green Chemistry*, 2026. 28 (29): 12428–12437. DOI: 10.1039/d6gc01450j
2. **Sidan Lu**, Allyson McGaughey, Sungju Im, Yiming Liu, Xinyi Wang, Aaron Leininger, David Jassby, Eric M. V. Hoek, and Zhiyong Jason Ren*. Membrane Electrolysis Distillation for Volatile Fatty Acids Extraction from pH-Neutral Fermented Wastewater. *Water Research*, 2024, 265, 122306. DOI: 10.1016/j.watres.2024.122306.
3. Aaron Leininger, **Sidan Lu**, Jinyue Jiang, Yanhong Bian, Harold D. May, and Zhiyong Jason Ren*. The Convergence of Lactic Acid Microbiomes and Metabolites in Long-Term Electrofermentation. *Environmental Science and Ecotechnology*, 2024, 22, 100459. DOI: 10.1016/j.esec.2024.100459.

4. **Sidan Lu**, Aaron Leininger, David Jassby, Eric M. V. Hoek, and Zhiyong Jason Ren*. Temperature Dependence of Acids Recovery from Wastewater Electro-Fermentation. *Resources, Conservation and Recycling*, 2024, 203, 107407. DOI: 10.1016/j.resconrec.2024.107407.
5. Sungju Im, Bongyeon Jung, Xinyi Wang, Jishan Wu, Minhao Xiao, Xin Chen, Javier A. Quezada-Renteria, Arpita Iddya, Derrick Dlamini, **Sidan Lu**, Christos T. Maravelias, Zhiyong Jason Ren, Eric M. V. Hoek*, and David Jassby*. High-Efficiency Recovery of Acetic Acid from Water Using Electroactive Gas-Stripping Membranes. *Environmental Science & Technology*, 2023, 57(27), 10096–10106. DOI: 10.1021/acs.est.3c01357.
6. Xinyi Wang, Sungju Im, Bongyeon Jung, Jishan Wu, Arpita Iddya, Javier A. Quezada-Renteria, Minhao Xiao, **Sidan Lu**, Jaewon Byun, Jeffrey Zhang, Zhiyong Jason Ren, Christos T. Maravelias, Eric M. V. Hoek*, and David Jassby*. Simple and Low-Cost Electroactive Membranes for Ammonia Recovery. *Environmental Science & Technology*, 2023, 57(25), 9405–9415. DOI: 10.1021/acs.est.3c01470.
7. **Sidan Lu**, Weiliang Sun, and Xiuping Zhu*. Synergistic Effects between Dual-Photoelectrodes and Bioanode Enhance Sustainable Hydrogen and Electricity Production from Wastewater. *Resources, Conservation and Recycling*, 2022, 183, 106367. DOI: 10.1016/j.resconrec.2022.106367.
8. **Sidan Lu**, Jun Lan, Weiliang Sun, Xiaojia He, and Xiuping Zhu*. High Energy Recovery from Salinity Gradients in a Concentration Flow Cell Enhanced by Bioelectrochemical Currents. *Chemical Engineering Journal*, 2021, 426, 130826. DOI: 10.1016/j.cej.2021.130826.
9. Alessandro Galia, Guangcai Tan, Massimo Marino, Federica Proietto, Onofrio Scialdone, **Sidan Lu**, and Xiuping Zhu*. Special Engines. *Salinity Gradient Heat Engines*, 2021, 302. DOI: 10.1016/B978-0-08-102847-6.00007-3.
10. **Sidan Lu**, Guangcai Tan, and Xiuping Zhu*. H₂ Evolution Catalysts for Microbial Electrolysis Cells. *Novel Catalyst Materials for Bioelectrochemical Systems: Fundamentals and Applications*, American Chemical Society, 2020, 27–43. DOI: 10.1021/bk-2020-1342.ch002.
11. **Sidan Lu**, Baiyun Lu, Guangcai Tan, William Moe, Wangwang Xu, Ying Wang, Defeng Xing, and Xiuping Zhu*. Mo₂N Nanobelt Cathodes for Efficient Hydrogen Production in Microbial Electrolysis Cells with Shaped Biofilm Microbiome. *Biosensors and Bioelectronics*, 2020, 167, 112491. DOI: 10.1016/j.bios.2020.112491.
12. Guangcai Tan, **Sidan Lu**, Nan Xu, Dingxue Gao, and Xiuping Zhu*. Pseudocapacitive Behaviors of Polypyrene-Grafted Activated Carbon and MnO₂ Electrodes to Enable Fast and Efficient Membrane-Free Capacitive Deionization. *Environmental Science & Technology*, 2020, 54(9). DOI: 10.1021/acs.est.9b07182.
13. **Sidan Lu**, Yujiao Sun*, Baiyun Lu, Danyang Zheng, and Shangwei Xu. Change of Abundance and Correlation of Nitrospira inopinata-Like Comammox and Populations in the Nitrogen Cycle during Different Seasons. *Chemosphere*, 2020, 241, 125098. DOI: 10.1016/j.chemosphere.2019.125098.

14. **Sidan Lu**, Binghan Xie, Bingfeng Liu, Baiyun Lu, and Defeng Xing*. Neglected Effects of Inoculum Preservation on the Start-Up of Psychrophilic Bioelectrochemical Systems and Shaping Bacterial Communities at Low Temperature. *Frontiers in Microbiology*, 2020, 10, 935. DOI: 10.3389/fmicb.2019.00935.
15. Guangcai Tan, **Sidan Lu**, Jizhou Fan, Guoqiang Li, and Xiuping Zhu*. Chloride-Ion Concentration Flow Cells for Efficient Salinity-Gradient Energy Recovery with Bismuth Oxychloride Electrodes. *Electrochimica Acta*, 2019, 322, 134724. DOI: 10.1016/j.electacta.2019.134724.
16. **Sidan Lu**, Hongna Li, Guangcai Tan, Fang Wen, Michael T. Flynn, and Xiuping Zhu*. Resource-Recovery Microbial Fuel Cells for Urine-Containing Wastewater Treatment without External Energy Consumption. *Chemical Engineering Journal*, 2019, 373, 1072–1080. DOI: 10.1016/j.cej.2019.05.130.
17. Guangcai Tan, Hongna Li, Haihui Zhu, **Sidan Lu**, Jizhou Fan, Guoqiang Li, and Xiuping Zhu*. Concentration Flow Cells Based on Chloride-Ion Extraction and Insertion with Metal Chloride Electrodes for Efficient Salinity-Gradient Energy Harvest. *ACS Sustainable Chemistry & Engineering*, 2018, 6(11), 15212–15218. DOI: 10.1021/acssuschemeng.8b03657.
18. Yujiao Sun*, **Sidan Lu**, Xuan Zhao, Aizhong Ding, and Lei Wang. Long-Term Oil Pollution and In Situ Microbial Response of Groundwater in Northwest China. *Archives of Environmental Contamination and Toxicology*, 2016, 72(4). DOI: 10.1007/s00244-017-0405-x.
19. **Sidan Lu**, Yujiao Sun*, Xuan Zha, Lei Wang, and Danyang Zheng. Impact of Precipitation on Fenghe River Water and Aquatic Microorganisms. *Environmental Science*, 2016. DOI: 10.13227/j.hjcx.2016.07.019.
20. **Sidan Lu**, Yujiao Sun*, Xuan Zhao, Lei Wang, Aizhong Ding, and Xiaohui Zhao. Sequencing Insights into Microbial Communities in the Water and Sediments of Fenghe River, China. *Archives of Environmental Contamination and Toxicology*, 2016, 71, 122–132. DOI: 10.1007/s00244-016-0277-5.
21. Danyang Zheng, Yujiao Sun*, Huijuan Li, **Sidan Lu**, Mingjun Shan, and Shangwei Xu. Multistage AO Activated-Sludge Process for Paraformaldehyde Wastewater Treatment and Microbial Community Structure Analysis. *Journal of Chemistry*, 2016. DOI: 10.1155/2016/2746715.

PRESENTATIONS

1. **Sidan Lu**, Amro Hassanein, Ji Hoon Lee, Ahmed Hemida, and Qilin Li*. Autonomous Modular Greywater Treatment System for Reliable Decentralized Water Reuse. YNOW Conference, New Orleans, Louisiana, 2026. *Invited talk*.
2. **Sidan Lu**, Hanno C. Erythropel, Paul T. Anastas, and Julie B. Zimmerman*. One-Pot Electrochemical Synthesis of a Nylon Precursor from Lignin-Derived Catechol. IWA-LET, Houston, Texas, 2026. *Poster presentation*.

3. Yunhao Zhang, **Sidan Lu**, Wanze Li, Qilin Li*. Overcoming Selectivity Bottlenecks in Decentralized Urine Resource Recovery: Novel Material Design for Nutrient-Targeted Electrodialysis. IWA-LET, Houston, Texas, 2026. *Poster presentation.*
4. **Sidan Lu**, Hanno C. Erythropel, Paul T. Anastas, and Julie B. Zimmerman*. One-Pot Electrochemical Synthesis of a Nylon Precursor from Lignin-Derived Catechol. ACS GCI, Boston, Massachusetts, 2025. *Poster presentation.*
5. **Sidan Lu**, Allyson McGaughey, David Jassby, Eric M. V. Hoek, and Zhiyong Jason Ren*. Membrane Electrolysis Distillation for Volatile Fatty Acid Separation from pH-Neutral Fermented Wastewater. ISMET, Houston, Texas, 2024. *Poster presentation.*
6. **Sidan Lu**, Aaron Leininger, David Jassby, Eric M. V. Hoek, and Zhiyong Jason Ren*. Temperature Dependence of Acid Recovery from Wastewater Electro-Fermentation. AEESP Research and Education Conference, Boston, Massachusetts, 2023. *Poster presentation.*
7. **Sidan Lu** and Xiuping Zhu*. Bioelectrochemical Systems for Resource Recovery and H₂ Production from Wastewater. 240th ECS Meeting, Orlando, Florida, 2021. *Contributed talk.*
8. **Sidan Lu**, Weiliang Sun, and Xiuping Zhu*. Solar-Biased Microbial Electrolysis Cells for Hydrogen Production under Visible-Light Irradiation. LSU Graduate Student Research Conference, 2021. *Contributed talk.*
9. **Sidan Lu**, Hongna Li, Guangcai Tan, and Xiuping Zhu*. Resource-Recovery Microbial Fuel Cells for Urine-Containing Wastewater Treatment without External Energy Consumption. ISMET7, Okinawa, Japan, 2019. *Poster presentation.*
10. **Sidan Lu**, Hongna Li, Guangcai Tan, and Xiuping Zhu*. Mo₂N Nanobelt Cathodes for Efficient Hydrogen Production in Microbial Electrolysis Cells. 4th IWA Conference of Science Summit on Urban Water, Harbin, China, 2019. *Poster presentation.*
11. **Sidan Lu**, Hongna Li, Guangcai Tan, Fang Wen, Michael T. Flynn, and Xiuping Zhu*. Resources Recovery from Wastewater Using Advanced Bioelectrochemical Systems. 4th International Young Scholars Shenzhou Forum, Harbin, China, 2019. *Contributed talk.*
12. **Sidan Lu**, Hongna Li, Guangcai Tan, Fang Wen, Michael T. Flynn, and Xiuping Zhu*. Resource-Recovery Microbial Fuel Cells for Space Wastewater Treatment. Louisiana Space Fall Council Meeting, Baton Rouge, Louisiana, 2017. *Poster presentation.*
13. **Sidan Lu**, Yujiao Sun*, Xuan Zhao, Aizhong Ding, and Lei Wang. Long-Term Oil Pollution and In Situ Microbial Response of Groundwater. IWA World Water Congress and Exhibition, Brisbane, Australia, 2016. *Contributed talk.*

TEACHING AND MENTORING EXPERIENCE

Course Assistant, Fall 2020 EVEG 3110 — Water and Wastewater Treatment Instructor: Dr. Xiuping Zhu, Louisiana State University

Assisted in preparing instructional materials and converting in-person classes to virtual classes.

Guest Lecture— Spring 2019 EVEG 4780: Renewable Energy and Power Generation Instructor: Dr. Xiuping Zhu, Louisiana State University

Delivered two lectures on electrochemical systems working mechanisms, energy generation abilities, and compared with existing technologies using renewable energy resources.

Guest Lecture– Fall 2019 CE 7105: Advanced Topics in Water Quality and Treatment Instructor: Dr. Xiuping Zhu, Louisiana State University

Delivered two lectures on working mechanisms of microbial fuel cell, microbial electrolysis cell, and concentration flow cell. Assigned student tasks to investigate advantages and disadvantages of each technology and followed up the progresses in the second lecture.

Course Assistant, Fall 2017 EVEG 3145: Environmental Engineering III—Water Chemistry

Instructor: Dr. Xiuping Zhu, Louisiana State University

Supported course delivery by preparing instructional materials, leading problem-solving sessions, and responding to student questions.

Science Outreach Mentor Assistant, Mentor: Dr. Xiuping Zhu, Louisiana State University

Middle school students, Caddo Middle Magnet School, Louisiana, 2019

Prepared different types of microbial fuel cells and assisted with the scientific demonstration to the students.

Science Outreach Mentor Assistant, Mentor: Dr. Xiuping Zhu, Louisiana State University

Middle school students, Science and Technology Charter School, Louisiana, 2018

Assisted with the scientific demonstration materials and answered student questions.

Mentoring and Research Advising

Donald Toledo, High school teacher, Research Experience for Teachers (RET) Program, Houston ISD 2026

David Ramos, Undergraduate researcher, Rice University 2026

Maritza Cisneros Estrada, Undergraduate researcher, Rice University 2026

Chan Kim, Graduate researcher, summer research program, Rice University 2026

Diego Delgado, Undergraduate researcher, Rice University 2026

Toby Jiang, Undergraduate researcher, Louisiana State University 2019–2020

Tristan Pluta, Undergraduate researcher, Louisiana State University 2018–2020

PROFESSIONAL SERVICE AND LEADERSHIP

Research Experience for Teachers Mentorship Program, Rice University 2026

Project Advisory Committee, Project 5371, The Water Research Foundation (WRF)

2025 – Present

Group Manager, Faculty-Postdoc Mentorship Program, Yale University 2025

Site Selection Committee, Association of Environmental Engineering and Science Professors (AEESP) 2024 – Present

Scientific Committee, National Science Foundation (NSF) 2024

Proposal Review Panel, 17th IWA World Congress on Anaerobic Digestion 2022

Vice president , Princeton University Chinese Student Christian Fellowship	2021– 2024
Vice president , Louisiana State University Chinese Student Christian Fellowship	2017– 2021

Reviewer: Environmental Science & Technology, Joule, Bioresource Technology, Frontiers in Microbiology, Science of the Total Environment, ACS Sustainable Chemistry & Engineering, Reaction Chemistry & Engineering, Environmental Pollution, and Environmental Science: Water Research & Technology

RESEARCH SKILLS

Chemical Analysis: Triple quadrupole mass spectrometry (QQQ-LC/MS), Gas chromatography–mass spectrometry (GC/MS), High-performance liquid chromatography (HPLC), Combustion ion chromatography (CIC), Nuclear magnetic resonance spectroscopy (NMR), and Inductively coupled plasma–optical emission spectroscopy (ICP-OES)

Materials and Biological Characterization: Scanning electron microscopy (SEM), X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), Three-dimensional X-ray computed tomography (3D X-ray CT), Energy-dispersive X-ray spectroscopy (EDX), DNA extraction, Next-generation 16S rRNA sequencing, PCR, and quantitative PCR (qPCR)

Microscopy: ZEISS ELYRA 7 with Lattice SIM² Super-Resolution Structured Illumination Microscopy and Olympus IX71 Inverted Fluorescence Microscopy

Modeling: Mass transfer, Chemical kinetics, and Chemical equilibrium